

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2287

Woodsdale 500 kV
August 25, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2287. The Interconnection Customer proposes a 786.1 MW natural gas combined cycle facility interconnecting at the Woodsdale 500 kV substation in the Ohio area. The request was received on February 14, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 564 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,896,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Goldsboro - Ironman 230kV is overloaded to 107.5% of its 617 MVA emergency rating for the loss of the Omni-Hackberry Substation - Norwich 500kV. The project contributes 6.9% DFAX (54.0 MW) to this overload.

The Queen Crk - Auglaize Hydro 230kV is overloaded to 105.4% of its 665 MVA emergency rating for the loss of the Wolfe - Barrett 500kV. The project contributes 6.9% DFAX (54.3 MW) to this overload.

For the loss of Barrett - Norwich 500kV, the Toledo Bend 500 kV/230 kV Transformer loads to 102.0% of its 818 MVA rating. Project DFAX: 8.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ironman - Goldsboro 230kV	Reconductor line	\$22,985,000	11% (\$2,526,000)
Auglaize Hydro - Queen Crk 230kV	Reconductor line	\$1,626,000	78% (\$1,274,000)
Toledo Bend 500 kV/230 kV Transformer	Replace transformer with higher-rated unit	\$23,393,000	13% (\$3,096,000)
	Total allocated to project		\$6,896,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.